

LECTURE 9

Culmination & Revision

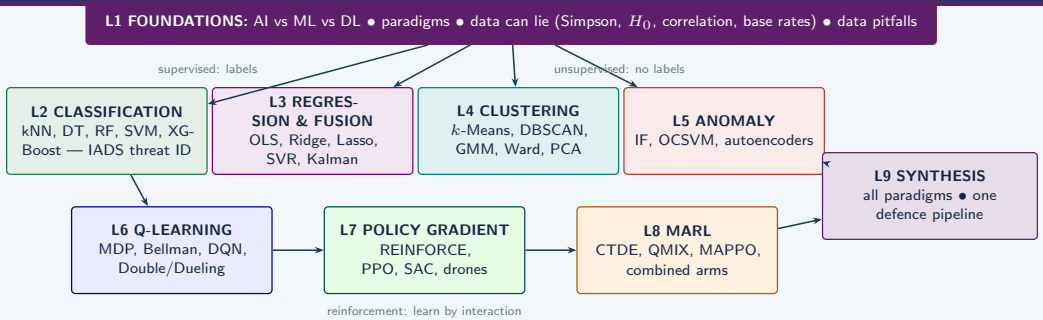
Integrating eight lectures into one coherent defence-AI framework
Concept map • case-study threads • algorithm selection • key formulas • misconceptions • ethics

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Course Concept Map — One Decision Tree, One Toolkit



Key thread: raw sensor data (L1 quality) → threat classification (L2–3) → pattern discovery (L4–5) → decision optimisation (L6–8). No algorithm exists in isolation.

Three master questions

① Do I have labels? (supervised vs unsupervised) ② Is the environment interactive? (static ML vs RL) ③ Are there multiple agents? (single-agent RL vs MARL)

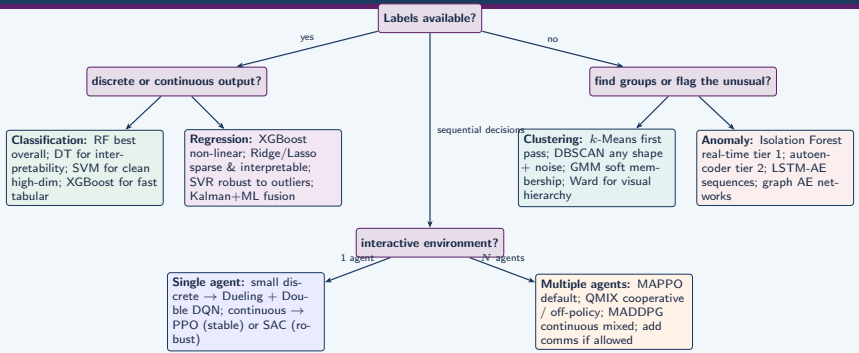
Cross-Lecture Case-Study Thread — One Battlefield, Eight Algorithms

L	Case study	Key result	Operational value
1	Data pitfalls (Afghan mine detector, Simpson, base rates)	94.1% Afghan $\rightarrow$ 58.7% Iraq	know when to trust your data
2	IADS 5-class threat classification (N=8,500)	RF 97.5%, AUC 0.998 (AB)	real-time threat ID for SAM crews
3	Mission duration + 3-sensor fusion	XGBoost R^2=0.912; KF+RF 96.2%	planning accuracy; robust fusion
4	OPFOR archetypes (N=8,000 trajectories)	k=5, silhouette 0.712	INFILTRATION found without labels
5	Network intrusion (500K) + insider threat	AE TPR 95.8%, FPR 0.6%; MTTD 31.4 h	zero-day detection; CERT ops
6	Wargame DQN (20×20 JANUS)	93% vs random; 81% vs novice	tactical advisory system
7	4-drone swarm, continuous control	PPO 94.3%; SAC 96.2% (91% real)	autonomous UAS coordination
8	6-unit combined-arms MARL	MAPPO 82.4% vs human 81.3%	emergent SEAD-first doctrine

L2: the PPTX revision table cites RF AUC 0.991; the L2 deck reports 0.998 for the AB one-vs-rest curve.

Golden thread: L4 discovered the INFILTRATION archetype $\rightarrow$ L5 flags such trajectories as anomalies $\rightarrow$ L6's wargame agent learns counter-infiltration positioning $\rightarrow$ L8's SEAD-first protocol denies infiltrators the SAM coverage gaps they exploit. The course is one ecosystem.

Algorithm Selection Guide — From Problem Statement to Method



Before any of these: apply Lecture 1's checks — collection bias, base rates, confounders, distribution shift. The best algorithm cannot rescue a biased dataset.

Best Performers by Task — the Course Scoreboard

Task	Best method	Result
Classification	Random Forest	97.5% (IADS)
Regression	XGBoost	R^2 0.912
Sensor fusion	Kalman + RF	96.2%
Imbalance	RF/XGB + SMOTE + weights	AB recall 89–92%
Clustering	k-Means (k=5)	silhouette 0.712
Anomaly (accuracy)	Autoencoder	TPR 95.8%
Anomaly (speed)	Isolation Forest	0.002 ms/packet

Task	Best method	Result
Discrete RL	Dueling + Double DQN	93% vs random
Continuous RL	SAC	96.2% (91% real)
Stable RL	PPO	94.3%, less tuning
Training aid	Curriculum	+40 points
Cooperative MARL	MAPPO	82.4%
MARL + comms	CommsMAPPO	88.1%
Comms-denied	MAPPO / QMIX	82.4% / 74.1%

Read every number with Lecture 1 in mind: all are single-study simulation or benchmark results without reported confidence intervals. They rank methods *for these case studies*; they are not guarantees for your data.

DATA & STATISTICS (L1)

$$\text{Error} = \text{Bias}^2 + \text{Variance} + \varepsilon, \quad P(\geq 1\ \text{FP}) = 1-(1-\alpha)^m$$

$$\text{PPV} = \frac{\text{sens} \cdot p}{\text{sens} \cdot p + (1 - \text{spec})(1-p)}$$

$$D_{KS} = \max_x |F_1 - F_2|$$

CLASSIFICATION (L2)

$$P = \tfrac{TP}{TP+FP},\ R = \tfrac{TP}{TP+FN},\ F_\beta = (1+\beta^2)\tfrac{PR}{\beta^2 P+R}$$

$$G = 1 - \sum p_k^2, \quad H = -\sum p_k \log_2 p_k$$

$$\text{margin} = 2/\|w\|$$

$$\min \tfrac{1}{2}\|w\|^2 + C\sum \xi_i, \quad K = e^{-\gamma\|x-x'\|^2}$$

REGRESSION (L3)

$$\hat{\beta} = (X^\top X)^{-1} X^\top y, \quad \hat{\beta}_{\text{ridge}} = (X^\top X + \lambda I)^{-1} X^\top y$$

$$\text{Lasso: } \min \|y - X\beta\|^2 + \lambda\|\beta\|_1, \quad R^2 = 1 - \tfrac{SS_{\text{res}}}{SS_{\text{tot}}}$$

KALMAN FILTER (L3)

$$\hat{x}^- = F\hat{x} + Bu, \quad P^- = FPF^\top + Q$$

$$K = P^- H^\top (HP^- H^\top + R)^{-1}$$

$$\hat{x} = \hat{x}^- + K(z - H\hat{x}^-), \quad P = (I - KH)P^-$$

Key Formulas (II) — Unsupervised Learning & Anomaly Detection (L4–L5)

CLUSTERING (L4)

$$J = \sum_{j} \sum_{x \in C_j} \|x - \mu_j\|^2, \quad \mu_j = \frac{1}{|C_j|} \sum_{x \in C_j} x$$

$$s(i) = \frac{b(i) - a(i)}{\max(a(i), b(i))} \in [-1, 1]$$

$$P(x) = \sum_{k} \pi_k \mathcal{N}(x; \mu_k, \Sigma_k), \quad r_{ik} = \frac{\pi_k \mathcal{N}_{ik}}{\sum_j \pi_j \mathcal{N}_{ij}}$$

$$C = V \Lambda V^\top, \quad Z = \bar{X} V_k, \quad \mathrm{EVR}_j = \lambda_j / \sum \lambda$$

DBSCAN: core if $|N_\varepsilon(p)| \geq$ MinPts; clusters are maximal density-connected sets; ARI measures agreement between two clusterings.

ANOMALY DETECTION (L5)

$$D_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}$$

$$s(x, \psi) = 2^{-E[h(x)]/c(\psi)}, \; c(\psi) = 2H(\psi - 1) - \frac{2(\psi - 1)}{\psi}$$

$$\mathrm{RE}(x) = \|x - f_{\mathrm{dec}}(f_{\mathrm{enc}}(x))\|^2$$

$$f(x) = \mathrm{sign}\Big(\sum \alpha_i K(x_i, x) - \rho \Big)$$

Key pitfall: the plain covariance Σ in Mahalanobis is contaminated by the anomalies themselves — use the MCD estimator (most central 50%).

Q-LEARNING (L6)

$$Q^* = \sum_{s'} P\big[R + \gamma \max_{a'} Q^*\big]$$

$$Q \leftarrow Q + \alpha\delta$$

$$\delta = r + \gamma \max Q' - Q$$

$$y = r + \gamma Q(s', a^*; \theta^-)$$

$$Q = V + A - \bar{A}$$

POLICY GRADIENT (L7)

$$\nabla J = \mathbb{E}[\nabla \log \pi \, G_t]$$

$$\hat{A} = r + \gamma V(s') - V(s)$$

$$r_t = \pi_{\text{new}}/\pi_{\text{old}}$$

$$L^{\text{CLIP}} = \mathbb{E}[\min(r_t \hat{A}, \text{clip}\,\hat{A})]$$

$$J_{\text{SAC}} = \mathbb{E}[r + \alpha H(\pi)]$$

MARL (L8)

$$\partial Q_{\text{tot}}/\partial Q_i > 0$$

$$L = \tfrac{1}{N} \sum_i L_i^{\text{CLIP}}$$

$$A_i = Q(s, \mathbf{a}) - b_i(s, \mathbf{a}_{-i})$$

$$b_i = \mathbb{E}_{a_i' \sim \pi_i} Q(s, (a_i', \mathbf{a}_{-i}))$$

ε-GREEDY AND ANNEALING

$$P(\text{greedy}) = 1 - \varepsilon + \tfrac{\varepsilon}{|A|}, \quad \varepsilon_t = \varepsilon_{\min} + (\varepsilon_{\max} - \varepsilon_{\min}) e^{-t/\tau}$$

SHAPING AND DISCOUNTING

$$F = \gamma\Phi(s') - \Phi(s), \qquad G_t = \sum_k \gamma^k r_{t+k+1}$$

Common Misconceptions — Exam Traps and Conceptual Errors (I)

"More data always helps." **False.** Biased data amplifies bias: more Afghan data would not fix Iraq (94.1% $\rightarrow$ 58.7%). Quality and domain match beat quantity.

"DBSCAN needs k." **False.** It discovers the number of clusters from density (ε, MinPts); k-Means and GMM need k.

"High accuracy = good model." **False.** On the real IADS stream (78.4% civilian) "always civilian" scores 78%+. Check per-class recall, AUC, confusion matrix.

"A baseline biases the gradient." **False.** $\mathbb{E}[b(s)\nabla \log \pi] = 0$ for any b not depending on a: less variance, zero bias.

"Percentages can be added or averaged." **False.** Simpson: A wins both stone sizes yet loses 78% vs 83% overall — totals come from counts.

"Replay is only for efficiency." **Incomplete.** It also breaks temporal correlation — without that, DQN often diverges.

"Not significant = no effect." **False.** The coating trial ($p = 0.072$, 40 sorties) was under-powered; 93 per arm were needed.

"t-SNE distances are meaningful." **False.** Only local neighbourhoods are preserved; never compare between-cluster distances.

Common Misconceptions (II) — More Traps

"QMIX just sums Q-values." **Misleading.** VDN sums; QMIX mixes non-linearly, conditioned on the global state through a hypernetwork, while keeping monotonicity.

"A 99%-accurate detector's alarms are 99% reliable." **False.** At a 0.1% base rate only ≈9% of alarms are real threats.

"SAC always needs more episodes than PPO." **Context-dependent.** SAC took more episodes to 70% on the drone task (1,500 vs 800), but reuses data via replay; choose by whether compute or environment interaction is scarce.

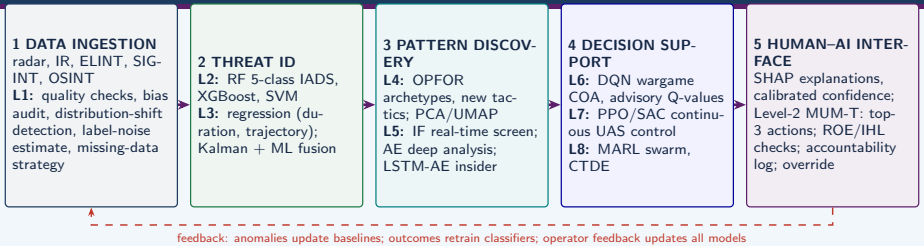
"Strong correlation shows the cause." **False.** Catering vs wins $r = 0.71$ fell to 0.09 after controlling for budget; maintenance hours vs breakdowns is reverse causation.

"Q = 18.7 means 18.7% chance to win." **False.** Q-values are expected discounted reward in reward units, not probabilities.

"Simulation win rate = field performance." **False.** Drones 94% → 78% real without randomisation; a human OPFOR adapts.

Exam technique: when a question offers a confident absolute ("always", "proves", "guarantees"), look for the hidden assumption — base rate, sample size, confounder, distribution shift, or simulation.

Integrated AI/ML Pipeline for Defence — End-to-End Architecture



Real-world caveat: the pipeline is ITERATIVE, not linear — every stage feeds back to earlier ones.

Deployment stack: stages 1–3 at the edge on FPGA/GPU (10–100 ms); stages 4–5 on servers with datalinks (100 ms–1 s); human decision 1–10 s. Total loop: **1.5–11 s** sensor-to-action.

Ethical AI in Defence — Eight Non-Negotiable Principles

1 Human control. Level 3+ unresolved under IHL; this course: Level 2 advisory. Meaningful control = understanding + oversight + ability to intervene.

2 Data integrity. L1 pitfalls flow into every decision: bias audits, shift monitoring, adversarial data tests.

3 Explainability. SHAP for classifiers, per-feature RE for anomalies, Q-value advisories — targeting needs a rationale.

4 Proportionality. IHL Art. 51(5)(b): no reward function can encode this legal, contextual judgement.

5 Distinction. Combatant vs civilian: novel uniforms, dual-use sites, surrender — no AI solves this reliably at tempo.

6 Adversarial robustness. Fixed policies can be reverse-engineered; classifiers fooled; red-team before deployment.

7 Accountability. Chain of command remains; full audit log: input $\rightarrow$ AI advice $\rightarrow$ human decision $\rightarrow$ outcome.

8 Sim-to-real honesty. 82.4% in simulation $\neq$ operational; drones 94% $\rightarrow$ 78%. Overstating AI is an ethical failure.

Ethical bright line: AI is a DECISION-SUPPORT tool. Human commanders retain authority for all lethal action.

Assessment Preparation — What Examiners Look For

Typical integration questions

- **Pipeline design:** "Design an AI cyber-defence pipeline for 10M packets/day" — expect L1 data audit + L5 3-tier + L2 classifier + L8 human control
- **Algorithm choice:** "QMIX or MAPPO for a naval task force?" — compare on cooperation, comms, action type, data efficiency
- **Failure analysis:** "DQN wins 93% vs random but loses to an adaptive OPFOR — why?" — fixed-policy exploitability; league self-play
- **Data interpretation:** "AI units win less overall" — spot Simpson's paradox and compute from counts
- **Ethics:** justify Level 2 over Level 3 using sim-to-real gaps and IHL

What distinguishes a top answer

- **Numbers, not adjectives:** "RF 97.5%, AB recall 96.1%, 2.8 ms" beats "RF is better"
- **Limitations stated:** sample size, confidence interval, simulation vs field, base rate
- **Cross-lecture links:** L4 $\rightarrow$ L5 $\rightarrow$ L6 $\rightarrow$ L8 thread
- **Human control:** autonomy level, accountability and IHL in every deployment answer
- **Worked arithmetic shown:** Kalman gain, silhouette, Q-update, PPV

Looking Forward — What This Course Equips You To Do

The three ideas that matter most

1. **Data quality and honest interpretation beat algorithm sophistication.** Simpson, base rates, survivorship and shift break more systems than weak models do.
2. **Method choice follows the problem:** labels? interaction? multiple agents? — then validate with several metrics.
3. **Simulation results need real-world caveats** and a human accountable for lethal decisions.

You can now

- Direct AI development teams with technical credibility
- Evaluate vendor and research claims with numerical rigour
- Challenge AI recommendations constructively and safely
- Design human–machine teaming protocols compatible with IHL

Next — Lecture 10: student project presentations and the final Q&A panel.